

# **BSBINS401 - Analyse and Present Research Information**

## **Session 2: Data Visualization Basics and Tools Overview**

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## Learning Objectives

- Understand different types of graphs and when to use them
- Familiarize with key data visualization tools and platforms
- Learn basic functionalities of Jupyter Notebooks and Kaggle
- Get an overview of Matplotlib and Seaborn libraries

# What is Data Visualization?

- **Definition:** The practice of translating information into a visual context
- **Purpose:**
  - Simplify the interpretation of complex data
  - Reveal patterns, trends, and correlations
  - Enhance data-driven decision making

## Key Graph Types and Their Uses

| Graph Type   | Use Case                           |
|--------------|------------------------------------|
| Line Chart   | Trend over time                    |
| Bar Chart    | Comparing discrete categories      |
| Scatter Plot | Relationship between two variables |
| Histogram    | Frequency distribution of data     |

# Tools for Data Visualization

- **Jupyter Notebooks**

- Interactive computing environment supporting live code, equations, visualizations, and narrative text
- [Jupyter Notebook Introduction Tutorial](#)

- **Kaggle**

- Platform for data science competitions and public datasets
- Includes an in-browser coding environment

- **Visualization Libraries**

- **Matplotlib**: Standard plotting library for Python ([Matplotlib Getting Started](#))
- **Seaborn**: High-level interface built on Matplotlib for statistical graphics ([Seaborn Introduction](#))

# Jupyter Notebook & Kaggle Overview

- **Jupyter Notebooks :**
  - Ideal for iterative development and data exploration
  - Combines code, output, and rich text in a single document
- **Kaggle :**
  - Create and share notebooks online
  - Explore datasets and participate in competitions
- **Demo :**
  - In-class demonstration of starting a notebook and navigating Kaggle

# Introduction to Matplotlib and Seaborn

- **Matplotlib:**
  - Widely used for creating static plots
  - Offers detailed control over every aspect of a figure
- **Seaborn:**
  - Simplifies creating attractive statistical graphics
  - Provides high-level interfaces for drawing appealing graphs

## Example Activity:

- Plot simple line charts and bar charts using sample code snippets

```
import matplotlib.pyplot as plt
import seaborn as sns

# Sample data
data = [1, 3, 2, 5, 7, 4]

# Matplotlib example
plt.plot(data)
plt.title("Simple Line Chart")
plt.show()

# Seaborn example
sns.barplot(x=list(range(len(data))), y=data)
plt.title("Simple Bar Chart")
plt.show()
```



## In-Class Activities

- **Explore Jupyter Notebooks and Kaggle :**
  - Create a new notebook and run a "Hello, World!" cell
  - Browse Kaggle to locate datasets and review notebook examples
- **Experiment:**
  - [Activity: Explore Different Types of Graphs](#)
  - Modify code snippets (colors, labels, etc.) to create your own visualizations

## Activities

- Activity: [Exploring Different Types of Graphs](#)
- Jupyter Notebook: [Introduction Tutorial](#)

## Lab Resources

- Lab: [Markdown in Jupyter Notebooks](#)
- Lab Exercise: [Simple Plotting with Matplotlib](#)

## Additional Resources

- **Matplotlib Documentation:** [Getting Started](#)
- **Seaborn Documentation:** [Introduction](#)
- **Kaggle Learn:** [Intro to Data Visualization](#)
- **Jupyter Notebook Tutorials:** [Jupyter Docs](#)

## Further Reading

- Article: [Data Visualization Best Practices](#)

## Summary & Next Steps

- **Today's Session:**
  - Reviewed key graph types, visualization tools, and platforms
  - Demonstrated usage of Jupyter Notebook and Kaggle
- **Next Session Preview:**
  - Dive deeper into data visualization techniques
  - Engage in more hands-on plotting exercises and explore advanced customization options

## Any Questions?

- Please ask any questions now or feel free to reach out during office hours.
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